

Assignment 3: Hypersurfaces and Black Hole Collapse

Deadline for Maths Students: Thursday, May 21st.

Deadline for Physics Students: Tuesday, May 26th.

Students taking the “[...] Black Holes” physics course are only expected to submit the exercises marked with an asterisk*, but are obviously invited to try all of them. Students taking the “[...] Classical Systems” maths course are expected to submit (almost) all exercises.

If you skip an exercise, be sure to read it at least, since you might need its result later.

We have hidden most hints as white text, to encourage you to try to solve the problems without looking at them. If you are stuck, simply copy the white text somewhere else to reveal it.

Feel free to use **Mathematica**, but submit the notebook or screenshots of it.

Exercise 1: The Projector

We defined the projector $h^{\mu\nu}$ on $\Sigma \subset \mathcal{M}$ as $h^{\mu\nu} = h^{ij} e_i^\mu e_j^\nu$, where $e_i^\mu = \frac{\partial x^\mu}{\partial y^i}$, fulfilling

$$g^{\mu\nu} = h^{\mu\nu} + \varepsilon n^\mu n^\nu, \quad (1)$$

where n_μ is the unit normal to Σ , with $\varepsilon = n^2$, fulfilling $e_i^\mu n_\mu = 0$ for all i .

a)* Prove

$$h_\nu^\mu = \delta_\nu^\mu - \varepsilon n^\mu n_\nu, \quad (2)$$

$$h^{\mu\nu} n_\mu = 0, \quad (3)$$

$$h_\rho^\mu h_\nu^\rho = h_\nu^\mu. \quad (4)$$

b) We also defined $e_\mu^i = h^{ij} g_{\mu\nu} e_j^\nu$. Prove

$$e_\mu^i e_j^\mu = \delta_j^i, \quad (5)$$

$$e_\mu^i e_i^\nu = h_\nu^\mu, \quad (6)$$

$$e_\mu^i n_\mu = 0. \quad (7)$$

Exercise 2: Extrinsic Curvature

We defined the extrinsic curvature as $K_{\mu\nu} = -h_\mu^\lambda \nabla_\lambda n_\nu$.

a)* Prove

$$K_{\mu\nu} = -h_\mu^\rho h_\nu^\sigma \nabla_\rho n_\sigma, \quad (8)$$

$$K_{\mu\nu} = n_\sigma h_\mu^\rho \nabla_\rho h_\nu^\sigma. \quad (9)$$

b) Writing $n = N df$, with $N = |g^{\mu\nu} \partial_\mu f \partial_\nu f|^{-1/2}$, prove that $K_{\mu\nu}$ is symmetric.

c)* Assuming that $K_{\mu\nu}$ is symmetric, prove

$$K_{\mu\nu} = -\frac{1}{2} \mathcal{L}_n h_{\mu\nu}. \quad (10)$$

Exercise 3: Oppenheimer-Snyder Collapse

In this exercise we will perform some necessary calculations for the Oppenheimer-Snyder collapse, which we will study in detail in class on Wednesday, May 13th.

In 1939, J. Robert Oppenheimer and his student Hartland Snyder published the first solution to the Einstein Equations describing a star's dynamical collapse to form a black hole.¹ They crudely modelled the collapsing star as a spherical ball of pressureless matter of uniform density.

Therefore, the metric inside the star is a FLRW metric with $k = 1$,

$$ds_-^2 = -d\tau^2 + a^2(\tau) [d\chi^2 + \sin^2(\chi)d\Omega^2] , \quad (11)$$

while the metric outside the star, due to Birkhoff's theorem,² must be the Schwarzschild metric,

$$ds_+^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2 , \quad (12)$$

where M is the star's total mass. Notice that the time coordinates t and τ are *not* the same.

To glue these two metrics along the surface of the star, we must first compute the induced metric and extrinsic curvature on the surface of the star, as seen from the two perspectives.

From the inside, the surface of the star is located at constant $\chi = \chi_0$ in comoving coordinates. From the outside, the surface of the star is at some $r = R(\tau)$ and $t = T(\tau)$, where τ is the proper time of observers at the surface of the star. Clearly this τ is the same as in (11).

a)* Choosing $y^i = (\tau, \theta, \phi)$ as coordinates on the surface of the star Σ , show that the induced metric, as seen from the inside, reads

$$ds_-^2|_\Sigma = -d\tau^2 + a^2(\tau) \sin^2 \chi_0 d\Omega^2 , \quad (13)$$

while, as seen from outside, it is

$$ds_+^2|_\Sigma = -\left(F\dot{T}^2 - F^{-1}\dot{R}^2\right) d\tau^2 + R^2 d\Omega^2 , \quad (14)$$

where $F = 1 - 2M/R$. *Hint:*

Since the induced metrics must match, this means

$$R(\tau) = a(\tau) \sin \chi_0 , \quad F\dot{T}^2 - F^{-1}\dot{R}^2 = 1 . \quad (15)$$

b)* Show that the unit normal vector, as seen from the inside, is

$$n^\mu = a^{-1} \delta_\chi^\mu . \quad (16)$$

Hint:

c)* Compute the extrinsic curvature, as seen from the inside,

$$K_{\tau\tau} = 0 , \quad K_{\theta\theta} = -a \sin \chi_0 \cos \chi_0 , \quad K_{\phi\phi} = -a \sin \chi_0 \cos \chi_0 \sin^2 \theta . \quad (17)$$

Hint:

¹Yes. They show this in Nolan's movie.

²In general relativity, Birkhoff's theorem states that any spherically symmetric solution of the vacuum Einstein Equations must be static. This means that the exterior solution (*i.e.* the spacetime outside of a spherical, non-rotating, gravitating body) must be uniquely given by the Schwarzschild metric. We will talk about this theorem in detail in future lectures.

d)* Show that the unit normal covector, as seen from the outside, is

$$n_\mu = (\dot{R}, -\dot{T}, 0, 0). \quad (18)$$

Hint:

e)* Compute the extrinsic curvature, as seen from the outside,

$$K_{\tau\tau} = n_\mu a^\mu, \quad (19)$$

where $a^\mu = u^\nu \nabla_\nu u^\mu$, and $u^\mu = e^\mu_\tau$ is the 4-velocity of an observer on the surface of the star (no need to compute a^μ explicitly), and

$$K_{\theta\theta} = FR\dot{T}, \quad K_{\phi\phi} = FR\dot{T} \sin^2 \theta. \quad (20)$$

You can use the Christoffel symbols for the Schwarzschild metric without deriving them.

Hint:

Exercise 4: Gaussian Normal Coordinates

In a D -dimensional manifold $\mathcal{M}$, consider a hypersurface $\Sigma = \{r(x^\mu) = 0\}$. *Gaussian normal coordinates* on $\mathcal{M}$ adapted to this hypersurface are defined by the conditions that $g^{\mu\nu} \partial_\mu r \partial_\nu r = 1$ and $g^{\mu\nu} \partial_\mu r \partial_\nu y^i = 0$, where the y^i are coordinates on Σ . It can be shown that these conditions always locally admit a solution.

a) Show that if we choose (r, y^i) as Gaussian normal coordinates on a neighbourhood of $\Sigma \subset \mathcal{M}$, then the metric can locally be written as

$$ds^2 = dr^2 + h_{ij}(r, y^k) dy^i dy^j. \quad (21)$$

b) Find the expression for the extrinsic curvature of Σ in these coordinates.

Exercise 5: ADM Decomposition

Consider a metric written in the $3+1$ form,

$$ds^2 = -N^2 dt^2 + h_{ij} (dx^i + N^i dt) (dx^j + N^j dt). \quad (22)$$

Let $T = \partial_t$, $X = N^i \partial_i$, and $n = (T - X)/N$.

a) Verify that $n_\mu = -N(dt)_\mu$ and hence n^μ is the future-directed unit normal vector to $\Sigma_t = \{t = \text{constant}\}$.

b) By writing n_μ in terms of T_μ and X_μ and using $K_{\mu\nu} = h^\rho_\mu h^\sigma_\nu \nabla_\rho n_\sigma$, show that

$$K_{ij} = \frac{1}{2N} (\partial_t h_{ij} - D_i N_j - D_j N_i), \quad (23)$$

where D_i is the Levi-Civita connection of h_{ij} , and $N_i = h_{ij} N^j$.

Exercise 6: Gauss' Theorema Egregium

Recall that we defined the Riemann tensor $\mathcal{R}^\mu_{\nu\rho\sigma}$ of the connection D_μ on $\Sigma \subset \mathcal{M}$ as

$$\mathcal{R}^\mu_{\nu\rho\sigma} V^\nu = 2D_{[\rho} D_{\sigma]} V^\mu, \quad (24)$$

for vectors V^μ tangent to $\Sigma \subset \mathcal{M}$, that is, for vectors fulfilling $V^\mu n_\mu = 0$.

a) *Extra exercise (not compulsory):* Prove Gauss' Equation,

$$h^\mu_\alpha h^\beta_\nu h^\gamma_\rho h^\delta_\sigma R^\alpha_{\beta\gamma\delta} = \mathcal{R}^\mu_{\nu\rho\sigma} - 2\varepsilon K_{[\rho}{}^\mu K_{\sigma]\nu}, \quad (25)$$

Warning: This is not a short computation. However, it is also not difficult. Simply expand the RHS of eq. (24) using the definition of D_μ and identify the relevant terms.

b) Contract Gauss' Equation to find the Contracted Gauss Equation,

$$R - 2\varepsilon n^\mu n^\nu R_{\mu\nu} = \mathcal{R} + \varepsilon(K^{\mu\nu} K_{\mu\nu} - K^2). \quad (26)$$

Hint:

Notice that in both cases, if our ambient spacetime $\mathcal{M}$ is flat, then $R^\alpha_{\beta\gamma\delta} = 0$, and so the intrinsic curvature $\mathcal{R}$ of the hypersurface Σ is directly related to its embedding in $\mathcal{M}$, that is, to its extrinsic curvature K .

c) Consider now a 2-surface embedded in $\mathbb{R}^3$. The two eigenvalues of $K_\mu{}^\nu$ are the principal values of the surface, $1/R_1$ and $1/R_2$.

Prove *Gauss' Theorema Egregium*: the product of the two principal radii $R_1 R_2$ is an *intrinsic* quantity, depending only on the induced metric on Σ and its (tangential) derivatives, but not on the normal vector and its derivatives.

Hint:

Exercise 7: Brane-Worlds

To explain our four-dimensional world with superstring theory, six spatial dimensions need to be hidden away. One way of doing so is by compactifying the extra dimensions. But we could also localize gravity on a lower-dimensional brane thanks to the curvature of the extra dimensions.

In this exercise, we will explore the Karch-Randall set-up, in which an AdS_5 bulk ends at an AdS_4 brane.³

Our starting point is the action

$$I = I_{\text{bulk}} + I_{\text{brane}}, \quad (27)$$

consisting of Einstein gravity in an AdS_5 bulk $\mathcal{M}$ ending on a brane $\partial\mathcal{M}$ with tension τ ,

$$I_{\text{bulk}} = \frac{1}{16\pi G_N} \left[\int_{\mathcal{M}} d^5x \sqrt{-g} (R[g] - 2\Lambda) - 2 \int_{\partial\mathcal{M}} d^4y \sqrt{-h} K \right], \quad (28)$$

$$I_{\text{brane}} = - \int_{\partial\mathcal{M}} d^4y \sqrt{-h} \tau, \quad (29)$$

where g denotes the bulk metric, h is the induced metric on $\partial\mathcal{M}$, G_N is the bulk Newton's constant, and Λ is the bulk cosmological constant, with curvature radius L .

³This model has recently gained renewed attention as a model to understand the Black Hole Information Problem, through what is known as "Brane-World Holography".

a) Not imposing the Dirichlet boundary condition on the brane, vary the action (27) to find the usual bulk Einstein Equations with a cosmological constant,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0, \quad (30)$$

plus the Israel Junction Condition on the brane,

$$K_{ij} - Kh_{ij} = 8\pi G_N \tau h_{ij}. \quad (31)$$

Warning: Do not compute everything from scratch, use all results seen in class.

b)* Our ansatz for the bulk geometry is

$$ds^2 = \frac{L^2}{\sin^2 z} (dz^2 + \hat{g}_{ij}(x) dy^i dy^j), \quad (32)$$

where $\hat{g}_{ij}$ is an AdS_4 metric with unit radius, and we place the brane at some constant $z = z_b$.

Compute the extrinsic curvature of the brane K_{ij} . Find the position of the brane as a function of the brane tension τ . What happens when $\tau \rightarrow 3/(8\pi G_N L)$?

Hint: